

BASSANO AGCM, AB, CanadaWMO: **710080**Lat: **50.8925N**Lon: **112.4650W**Elev: **807**StdP: **92.00**Time Zone: **-7.00 (NAM)**Period: **07-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-31.2	-27.9	-34.7	0.2	-31.2	-31.1	0.2	-27.7	15.6	-7.1	12.9	-2.7	2.2	280	0.720

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.2	30.9	16.4	28.8	15.8	26.9	15.5	18.3	26.1	17.4	25.6	16.6	24.6	4.8	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.6	12.2	21.4	14.5	11.4	20.4	13.5	10.6	19.3	55.0	26.1	52.2	25.9	49.5	24.6	23.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.3	11.4	9.8	-36.2	34.5	2.9	2.1	-38.3	36.1	-40.0	37.3	-41.6	38.5	-43.6	40.1
			-36.3	21.0	2.8	1.2	-38.4	21.8	-40.0	22.6	-41.6	23.3	-43.7	24.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.7	-9.3	-10.4	-3.2	4.0	10.7	14.8	17.7	17.0	12.1	4.8	-3.5	-10.8
	DBStd	12.05	8.50	8.46	8.21	5.07	4.28	2.97	2.66	3.11	4.65	5.07	7.65	8.15
	HDD10.0	3063	598	570	410	184	44	2	0	1	30	173	407	644
	HDD18.3	5382	856	803	668	429	237	112	44	63	191	421	656	903
	CDD10.0	779	0	0	0	6	66	146	239	218	93	11	1	0
	CDD18.3	57	0	0	0	0	1	5	24	22	4	0	0	0
	CDH23.3	1526	0	0	0	7	98	161	542	517	195	7	0	0
	CDH26.7	461	0	0	0	1	17	33	172	176	61	1	0	0
(14) Wind	WSAvg	4.2	4.4	3.8	4.2	4.7	4.7	4.4	3.8	3.7	3.9	4.1	4.1	3.9
(15) Precipitation	PrecAvg	350	12	11	17	30	53	71	43	39	29	17	16	12
	PrecMax	534	26	24	57	68	92	201	102	94	110	47	38	28
	PrecMin	159	1	1	1	8	13	23	7	1	2	3	1	1
	PrecStd	67	6	6	11	15	22	38	26	22	23	11	10	7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.9	9.3	16.9	23.4	28.7	29.9	33.3	33.7	31.4	24.0	17.9	9.7
		MCWB	3.9	4.2	8.1	10.9	13.4	17.6	18.0	16.3	15.2	12.4	9.1	4.6
	2%	DB	6.2	6.0	13.9	20.0	26.1	27.2	30.8	30.8	28.6	19.8	13.2	5.5
		MCWB	2.7	2.2	6.1	9.4	12.8	15.3	17.1	16.3	14.4	10.4	6.4	2.2
	5%	DB	4.3	3.5	10.7	16.8	23.5	25.0	28.5	28.5	25.3	17.0	9.6	3.3
		MCWB	1.6	1.0	4.7	7.4	11.8	14.5	16.8	16.0	13.4	9.1	4.5	0.8
	10%	DB	2.5	1.2	7.4	13.8	20.8	23.0	26.4	26.2	22.4	14.1	6.5	0.7
		MCWB	0.5	-0.5	3.0	5.8	10.6	13.6	16.4	15.5	12.5	7.4	2.4	-1.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.7	5.1	8.7	11.3	14.5	18.5	20.6	19.0	16.4	12.8	9.2	5.2
		MCDB	8.2	9.0	15.9	22.3	26.6	26.3	27.7	26.6	27.3	21.8	17.8	9.3
	2%	WB	2.8	2.4	6.7	9.4	13.3	16.9	18.8	17.8	15.1	11.0	6.7	2.4
		MCDB	5.7	5.3	12.8	19.1	24.3	24.3	26.0	26.4	25.7	18.3	12.4	5.3
	5%	WB	1.7	1.1	4.9	7.7	12.3	15.6	17.8	17.0	13.9	9.3	4.8	0.8
		MCDB	4.2	3.5	10.2	15.6	21.8	22.7	25.5	26.0	24.0	16.1	9.2	3.3
	10%	WB	0.4	-0.5	3.1	6.3	11.2	14.6	17.0	16.1	12.7	7.8	2.8	-1.1
		MCDB	2.5	1.1	7.1	12.9	19.2	21.4	24.9	24.2	21.7	13.6	5.9	0.7
(35) Mean Daily Temperature Range	5% DB	MDBR	10.8	11.6	12.1	13.9	14.7	13.6	15.2	15.6	15.6	14.3	12.0	10.8
		MCDBR	11.5	13.5	16.6	20.3	19.7	17.6	19.8	21.1	21.7	19.0	15.8	12.9
		MCWBR	8.7	10.3	9.9	10.6	8.6	7.8	8.2	8.2	9.5	10.3	9.9	10.0
		MCDBR	11.1	13.2	15.7	19.5	17.2	15.9	16.7	17.4	19.7	18.2	15.7	13.0
	5% WB	MCWBR	8.5	10.1	9.8	10.4	7.8	7.7	7.8	7.7	9.0	10.2	10.1	10.3
(40) Clear-Sky Solar Irradiance	taub		0.267	0.274	0.292	0.345	0.370	0.360	0.359	0.377	0.330	0.288	0.259	0.246
	taud		2.388	2.445	2.458	2.309	2.291	2.368	2.399	2.324	2.432	2.527	2.490	2.414
	Ebn at Noon		778	877	921	895	885	895	890	852	863	850	793	755
	Edh at Noon		60	74	89	116	124	116	111	114	91	68	54	50
(44) All-Sky Solar Radiation	RadAvg		1.17	2.17	3.52	4.80	5.76	6.25	6.75	5.42	3.83	2.34	1.27	0.89
	RadStd		0.09	0.20	0.27	0.35	0.40	0.39	0.33	0.36	0.37	0.27	0.11	0.08

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (83 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page